Jinyuan Yu

Research Assistant in 3D Reconstruction and Neural Rendering | CEO, 深圳市视元引擎有限公司

Seeking research and industry collaborators in 3D vision, neural rendering, computer graphics, and avatar reconstruction

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My research focuses on 3D/4D computer vision and geometry-aware generative modeling, especially monocular and sparse-view reconstruction, 3D Gaussian Splatting, neural rendering, depth estimation, novel-view synthesis, and human avatars. I build complete research systems spanning data preparation, geometry, PyTorch training, evaluation, GPU inference, and interactive visualization. I also lead a company developing low-cost 2D-to-3D conversion, 3D content generation, and hardware-interoperable glasses-free 3D display systems.

Research Interests

3D Gaussian Splatting; neural rendering; monocular and sparse-view 3D reconstruction; novel-view synthesis; depth estimation; occlusion-aware completion; human avatars; temporal image reconstruction.

Education and Appointments

Present	CEO, 深圳市视元引擎有限公司
2025-2026	Research Assistant, Department of Electrical and Electronic Engineering, Southern University of Science and Technology
2022-2025	M.S. in Computer Technology, Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences
2015-2019	B.S. in Computer Science and Technology, Shandong University of Science and Technology

Selected Research Experience

DepthWarpVS: Real-Time Multi-View Synthesis for Glasses-Free 3D Communication

- Designed depth-based forward warping and the hole/valid/pollute reliability representation for localized completion.
- Developed the lightweight MGMI completion network and a geometry-driven synthetic defect-generation pipeline.
- Measured approximately 5 ms per 4K target-view completion and approximately 15 FPS for 11-view output on one RTX 4090 in the current prototype.
- Related work accepted as a co-authored paper at ICDT 2026.

FlashAvatar-DepthFusion: Monocular Head Avatar Reconstruction

- Extended the official FlashAvatar pipeline with aligned per-frame depth and a scale-invariant Pearson-correlation loss.
- Designed conservative eroded face masks that exclude mouth interiors and uncertain silhouette regions.
- Built preprocessing validation and qualitative cross-view comparison tools for three identities; quantitative benchmarking remains ongoing.

VDA Absolute Depth Distillation

- Adapted Video Depth Anything for single-image feature extraction and trained a lightweight scale/shift prediction head with a frozen backbone.
- Implemented reciprocal-affine calibration using Depth Pro pseudo-labels, alignment, training, inference, and temporal comparison tools.
- The 1,800-frame in-scene snapshot reports teacher-referenced MSE of 0.1294 and 0.2248 on two held-out cameras; cross-scene evaluation is not yet complete.

MALA: Temporal-Spatial Remote-Sensing Reconstruction

- Developed EMAE and the MALA extension for cloud, strip, and mixed missing patterns in satellite image sequences.
- Combined temporal attention, multi-scale spatial features, explicit masks, and optional LaMA initialization.
- Across four reported representative cases, MALA averages 31.05 dB PSNR versus 28.89 dB for EMAE; paper under review and two related patent applications.

Knee MRI 3D Communication

- Built a privacy-minimized MRI package and export pipeline, WebGL2 volume/cutaway viewer, segmentation interfaces, and MRI-derived Gaussian delivery path.
- Delivered one four-sequence real MRI derivative case and a 47,443-Gaussian browser prototype with automated Python and Node tests.
- The project is an engineering prototype and does not claim clinical validation.

Publications and Research Outputs

- DWvs: Depth-Guided Image Warping and Hole Filling for Novel View Synthesis. Co-author, ICDT 2026, accepted.
- Research on Spectral Interpolation and Extrapolation of Remote Sensing Images Based on Temporal Generation Model. Under review.
- Patent application CN202511188342.5: temporal-stability partitioning and Transformer-LaMA interpolation.
- Patent application 202511314521.9: masked-autoencoding interpolation for sea-surface reflectance products.
- Patent application CN202311554601.2: multi-vehicle urban sensing optimization.

Technical Skills

Research: 3D Gaussian Splatting, neural rendering, novel-view synthesis, depth estimation, temporal modeling, image/video restoration
Programming: Python, C++, MATLAB, R, SQL
Frameworks and systems: PyTorch, CUDA/GPU experimentation, Linux, Git, Docker, Kubernetes, CPLEX, Gurobi

Selected Software

- MALA - https://github.com/jinyuanyu/MALA
- DepthWarpVS - https://github.com/jinyuanyu/depth_warp_vs
- VDA Absolute Depth Distillation - https://github.com/jinyuanyu/VDA-Absolute-Depth-Distillation
- FlashAvatar-DepthFusion - https://github.com/jinyuanyu/FlashAvatar-DepthFusion
- Knee MRI 3D Communication - https://github.com/jinyuanyu/knee-mri-3d-communication